

AI Productivity Agent: Architecture & Requirements Document

1. Executive Summary & Purpose

Agent Name: SyncBot

Mission: An intelligent autonomous productivity assistant designed to help professionals reclaim 5-10 hours weekly by automating calendar management, meeting scheduling, task extraction, and contextual information retrieval.

2. Target Audience & User Persona

- **Primary Persona:** Sarah Jenkins (32, Senior Product Manager)
- **Pain Points:** 2+ hours daily spent in manual calendar coordination, lost action items from meetings, and fragmented tool context across Slack, Notion, and Google Calendar.
- **Goals:** Seamless multi-participant scheduling, automatic task extraction from raw meeting transcripts, and reliable daily briefings.

3. Core Architectural Modules

Module	Technology	Responsibility
Perception Layer	FastAPI + Webhooks	Captures inbound Slack messages, email requests, and calendar invites
Reasoning Core	GPT-3.5-Turbo / ReAct Agent	Interprets intent, extracts dates/participants, and formulates step-by-step plans
Memory Subsystem	LangChain ConversationBuffer	Maintains user preferences (e.g., "No morning meetings", "Prefer 30-min syncs")
Tool Layer	Google Calendar API + Jira API	Performs slot availability checks, invite generation, and ticket creation
Responsible AI Filter	Presidio PII Masker + Guardrails	Strips sensitive employee information before LLM ingestion

4. Key Use Cases & Execution Flows

1. **Multi-Participant Scheduling:** Find overlapping free slots across 3+ internal and external calendars, compose polite invite emails, and dispatch Google Calendar invites automatically.
2. **Action Item Extraction:** Ingest raw audio transcripts or messy meeting notes, identify assignees and deadlines, and post structured tasks to Notion/Jira.
3. **Daily Executive Briefing:** Provide a conversational summary at 8:30 AM highlighting top 3 priorities, first meeting context, and pending follow-ups.
4. **Emergency Rescheduling Cascade:** Clear an entire afternoon calendar block with one command, notifying all participants with apology notes and automated alternative time slots.